

AHMAD ARRABI

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RESEARCH OVERVIEW

I am a 4th-year Ph.D. candidate in Computer Science at the University of Vermont. My research intersects diffusion models, multimodal generation, and autonomous medical imaging systems, where I develop models that enable intelligent positioning of intraoperative X-ray systems. I am particularly interested in applications in image-guided interventions and multi-modal learning.

EDUCATION

Ph.D. in Computer Science

2023 – 2027 (expected)

University of Vermont

Advisor: Safwan Wshah

B.S. in Computer Engineering

2017 – 2022

Princess Sumaya University for Technology

RESEARCH EXPERIENCE

Graduate Research Assistant

2023 – Present

University of Vermont – Vermont Artificial Intelligence Laboratory (VaiL)

- Routinely execute large-scale experiments on HPC clusters (Slurm), managing multi-GPU training pipelines
- Drove projects end-to-end from problem formulation and data curation to open-source release and publication

Medical Imaging

- Collaborated with interventional radiologists at the Cleveland Clinic and UVMHC to translate procedural C-arm positioning requirements into deep learning-based solutions for autonomous localization
- Led the development of a self-supervised framework for autonomous C-arm localization from intraoperative X-rays
- Designed an uncertainty-aware control pipeline for C-arm servoing toward predefined anatomical targets
- Fine-tuned multimodal large language models for skeletal landmark localization and integrated them into an agentic C-arm control pipeline for spatial reasoning and target-driven navigation

Generative Models (Diffusion)

- Designed a diffusion bridge framework for joint multimodal generation through controllable transport between diffusion trajectories during sampling
- Adapted large-scale diffusion and vision-language foundation models for task-specific objectives, including interleaved co-generation, controllable image editing, and multimodal synthesis
- Developed geometry-guided conditioning strategies for cross-view diffusion under extreme viewpoint shifts, reducing domain gap in ground-to-aerial generation

Undergraduate Research Assistant

2021

Princess Sumaya University for Technology

- Co-authored peer-reviewed papers on machine learning and reinforcement learning for autonomous driving and medical applications

PUBLICATIONS

Conferences

- **Arrabi, A.**, Jung, J., Le, J., Nguyen, A., Reed, J., Stahl, E., Franssen, N., Raymond, S., & Wshah, S. (2025). C-ARM Guidance: A Self-Supervised Approach to Automated Positioning During Stroke Thrombectomy. In 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI) (pp. 1-4)
- **Arrabi, A.**, Zhang, X., Sultani, W., Chen, C., & Wshah, S. (2025). Cross-View Meets Diffusion: Aerial Image Synthesis with Geometry and Text Guidance. In 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (pp. 5356-5366)

- Jung, J. H., **Arrabi, A.**, Luo, J., Scott, R., & Wshah, S. (2026). Autonomous Skeletal Landmark Localization towards Agentic C-Arm Control. Information Processing in Computer-Assisted Interventions (IPCAI), **Accepted**
- Lehyeh, A. A., Zhang, X., **Arrabi, A.**, Sultani, W., Chen, C., & Wshah, S. (2026). GeoFlow: Real-Time Fine-Grained Cross-View Geolocalization via Iterative Flow Prediction. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **Accepted**

Peer-Reviewed Workshops

- **Arrabi, A.**, Jung, J., Luo, J., Franssen, N., Raymond, S., & Wshah, S. (2025). Automated C-Arm Positioning via Conformal Landmark Localization. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops (pp. 4392-4401)
- Chen, Y., Zhang, X., **Arrabi, A.**, Sultani, W., Chen, C., & Wshah, S. (2026). Turning Generators into Retrievers: Unlocking MLLMs for Natural Language-Guided Geo-Localization. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings Workshops, **Accepted**

Journals

- Al-Mousa, A., **Arrabi, A.**, & Daoud, H. (2025). A reinforcement learning-based reverse-parking system for autonomous vehicles. IET Intelligent Transport Systems, 19(1), e12614

Abstracts

- Kashkoush A., **Arrabi A.**, Jung J., Franssen N., Wshah S., Raymond S. (2025). Deep learning model predicts biplane trajectory from pelvis to head. Society for NeuroInterventional Surgery, 7/2025.

ACADEMIC SERVICES & AWARDS

Reviewing:

- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.
- International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2026.
- International Journal of Computer Assisted Radiology and Surgery (IJCARS)
- Pattern Recognition
- Pattern Recognition Letters
- European Conference on Artificial Intelligence (ECAI), 2025.

Invited Talks:

- International Workshop on Video Surveillance Systems in Smart Cities, WACV 2025.

Awards:

- MICCAI Society Membership Grant, 2026.